

1 Model Formulation

$1, 2, \dots, K$ arms, $1, 2, \dots, T$ Time Horizon

2 Supplementary Material for Code

- `src/adversarial.py` <function object> `gen_adv_seq` : Refer to A1
- `src/hybrid.py` <class object> `gauss_hybrid_model` : Refer to A2
- `src/algorithms/EXP3.py` <class object> `EXP3_policy` : Refer to A3

3 A1

Algorithm 1 Generate Adversarial Sequence

Require: Horizon T , Autocorrelation Parameter c , Observation Ratio ρ

Ensure: Sequence $X_1, X_2, \dots, X_T$ where $X_i \in \{0, 1\}$, $\frac{1}{T} \sum_{t=1}^T X_t = \rho$

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1:  $X_1 \sim \text{Bernoulli}(\rho)$ 
2: for  $t = 2, \dots, T$  do
3:    $\hat{p}_t \leftarrow \frac{1}{t-1} \sum_{s=1}^{t-1} X_s$   $\triangleright$  Running empirical mean
4:    $e_t \leftarrow \hat{p}_t - \rho$   $\triangleright$  Error from target
5:   if  $X_{t-1} = 1$  then
6:      $p_{\text{stay}} \leftarrow \frac{1+c}{2} - e_t$ 
7:   else
8:      $p_{\text{stay}} \leftarrow \frac{1+c}{2} + e_t$ 
9:    $p_{\text{stay}} \leftarrow \text{clip}(p_{\text{stay}}, 0, 1)$ 
10:   $\text{stay} \sim \text{Bernoulli}(p_{\text{stay}})$ 
11:  if  $\text{stay}$  then
12:     $X_t \leftarrow X_{t-1}$ 
13:  else
14:     $X_t \leftarrow 1 - X_{t-1}$ 
15: return  $X_1, \dots, X_T$ 
```
